

README_TableTask_Paper-Based_Reliability_Analysis_EG

TableTask within-person reliability analysis based on Dudarev et al. (2023)

This document describes the purpose, workflow, statistical model, inputs, outputs, and interpretation of 03b_script_tabletask_within_person_reliability_paper_method_EG.R. The script adapts the time-sensitive and random split-half reliability framework from Dudarev et al. (2023) and the authors' released R code to the TableTask project.

1. Purpose and Scope

The purpose of Script 03b is to estimate study-level within-person reliability of heart-rate measurements across repeated TableTask trials. It evaluates whether two closely timed, split-half summaries of the same participant-trial produce consistent values after stable between-participant differences are accounted for with a mixed-effects model.

Script 03b is separate from Script 03a. Script 03a provides a trial-level odd-even RR consistency measure for quality-control purposes. Script 03b implements the paper-based split-half mixed-model analysis and produces reliability estimates for the dataset or defined analysis strata.

2. Relationship to the Published Method

The published framework distinguishes between between-person and within-person reliability. Script 03b implements the within-person component. The paper's time-sensitive method orders measurements by acquisition time, divides them into odd- and even-ordered observations within each participant and repeated situation, averages each half, and predicts one half from the other using a linear mixed-effects model.

Component	Published workflow	TableTask adaptation
Physiological measure	Repeated HR or HRV measurements	Step 3 aligned 4 Hz HR in BPM
Participant identifier	Individual participant	Participant A or participant B
Repeated situation	Day, activity, or another repeated context	One participant's one TableTask trial
Time-sensitive split	Chronological odd and even observations	Chronological odd and even 4 Hz HR samples
Reliability model	One split-half mean predicted from the other	$\text{odd_half_mean} \sim \text{even_half_mean} + (1 \mid \text{participant})$
Reliability estimate	Fixed-effect slope beta	Fixed-effect slope beta
Sensitivity analysis	Random split repeated 1,000 times	Random split repeated 1,000 times by default

3. Input Data

Required input directory:

processed_ecg_hr/trial_segments_4hz/paired_trial_hr/

Each paired trial CSV must contain:

- TimeRelTrialSec
- HR_A_BPM
- HR_B_BPM

The script also reads available identifiers and metadata, including:

- participant_A and participant_B
- sensor_A and sensor_B, used only as technical fallbacks if participant identifiers are absent
- recording_folder and dyad_id
- candidate_window, trial, and condition
- practice and pilot flags
- main-analysis exclusion flag and exclusion reason

4. Analysis Workflow

- Locate all paired Step 3 trial CSV files.
- Read participant A and participant B HR series from each file.
- Sort observations chronologically using TimeRelTrialSec.
- Reshape each dyad file into participant-level long-format data.
- Retain finite, positive HR observations only.
- Define one participant-trial as one repeated situation.
- Combine all participant-trial observations across the dataset.
- Create the planned analysis strata.
- For the time-sensitive analysis, divide chronological observations into odd- and even-ordered samples within each participant-trial.
- Calculate one odd-half mean and one even-half mean for every participant-trial.
- Fit a linear mixed-effects model with participant as a random intercept.
- Use the fixed-effect slope beta as the time-sensitive within-person reliability estimate.
- Repeat the split-half procedure with random assignment 1,000 times.
- Fit the same mixed model for every random split.
- Summarize the distribution of 1,000 random-split beta estimates.
- Export model summaries, split-half tables, random-beta tables, diagnostics, plots, and run documentation.

5. Statistical Model

The script fits the following linear mixed-effects model:

$$\text{odd_half_mean} \sim \text{even_half_mean} + (1 \mid \text{participant})$$

The dependent variable is the mean of the odd-ordered observations. The fixed-effect predictor is the mean of the even-ordered observations. The participant random intercept accounts for repeated participant-trial observations contributed by the same participant.

The fixed-effect slope for even_half_mean is interpreted as the within-person reliability estimate. The model is fitted with `lmerTest::lmer()`, which provides Satterthwaite-adjusted degrees of freedom and p-values.

6. Time-Sensitive Split-Half Analysis

- Observations are ordered by acquisition time within each participant-trial.
- The first, third, fifth, and subsequent odd-positioned observations form the odd half.
- The second, fourth, sixth, and subsequent even-positioned observations form the even half.
- Each half is averaged to produce two closely timed estimates for the same participant-trial.
- A participant-trial is retained only when both halves contain at least two observations.
- The time-sensitive beta is the primary paper-style reliability estimate.

7. Random Split-Half Sensitivity Analysis

- Observations are randomly reordered within each participant-trial.
- The randomized observations are divided into two approximately equal halves.
- The mean of each half is calculated.
- The same mixed-effects model is fitted.
- The process is repeated 1,000 times by default.
- The script reports the mean, standard deviation, median, 2.5th percentile, and 97.5th percentile of the successful beta estimates.
- A histogram is created for each analysis stratum, with the time-sensitive beta marked by a dashed vertical line.

8. Analysis Strata

The script reports reliability separately for:

- all available data
- all non-practice experimental trials
- main-analysis-eligible trials
- main-analysis-eligible FF trials
- main-analysis-eligible FE trials
- all available practice trials

These strata allow the study team to compare the full dataset with the planned main-analysis subset. The script does not automatically decide which stratum should be reported as the primary manuscript result.

9. Output Structure

processed_ecg_hr/paper_based_within_person_reliability_EG/
 tables/
 plots/
 README_03b_reliability_outputs_EG.txt

10. Main Output Files

File	Purpose
03b_within_person_reliability_summary_EG.csv	Primary reliability summary for all analysis strata.
03b_reliability_file_manifest_EG.csv	Lists successful and failed input files.
03b_reliability_long_input_EG.csv	Participant-level long-format HR data used by the analysis.

03b_time_sensitive_split_halves_<stratum>_EG.csv	Odd- and even-half means for each participant-trial.
03b_time_sensitive_model_<stratum>_EG.txt	Full mixed-model summary for the time-sensitive analysis.
03b_random_split_betas_<stratum>_EG.csv	Beta estimate from each random split iteration.
03b_random_split_beta_distribution_<stratum>_EG.png	Histogram of random-split beta estimates.
03b_reliability_run_summary_EG.txt	Execution settings and dataset counts.

11. Interpretation

A larger positive beta indicates stronger agreement between the two split-half estimates after participant-level differences are modeled. A beta near 1 indicates that participant-trial values from one half are closely reproduced by the other half. Lower beta values indicate weaker consistency.

Beta is a mixed-model slope, not a Pearson correlation coefficient, ICC, or Cronbach's alpha. The p-value tests whether the fixed-effect slope differs from zero; it does not by itself establish that the reliability is sufficiently high for a particular scientific purpose.

12. Important Statistical Considerations

- The script implements the paper's split-half mixed-model framework, not a trial-level QC correlation.
- The TableTask adaptation treats one participant-trial as the repeated situation.
- The current measurement unit is the interpolated 4 Hz HR sample.
- Adjacent 4 Hz samples are strongly autocorrelated and partly determined by interpolation; therefore, reliability may be very high.
- The resulting beta should be interpreted as consistency of the processed 4 Hz HR series, not as independent validation of raw ECG or R-peak accuracy.
- The study team should confirm whether 4 Hz samples or coarser non-overlapping time-bin summaries are the intended measurement units.
- FF and FE estimates should be compared descriptively unless a separate formal model is specified for comparing reliability across conditions.
- Singular mixed-model fits are flagged and should be reviewed before reporting results.

13. Relationship to Script 03a

Feature	Script 03a	Script 03b
Purpose	Trial-level QC	Study-level reliability
Input	Corrected RR intervals	Paired 4 Hz HR trial series
Statistic	Odd-even RR Pearson correlation	Linear mixed-effects slope beta
Output level	Sensor/trial and dyad/trial	Dataset or analysis stratum
Use	Flag locally unstable trials	Evaluate within-person measurement consistency

14. Software Requirements

Required R packages:

- readr
- dplyr
- lmerTest

```
install.packages(c("readr", "dplyr", "lmerTest"))
```

15. Running the Script

```
cd /Users/elnaz/Desktop/veronica_project/TableTask
```

```
Rscript 03b_script_tabletask_within_person_reliability_paper_method_EG.R \
--root . \
--random-iterations 1000 \
--overwrite
```

16. Recommended Reporting Language

Within-person reliability of processed heart-rate measurements was estimated using a time-sensitive split-half mixed-effects approach adapted from Dudarev et al. (2023). Within each participant and trial, chronologically ordered HR observations were divided into odd and even samples and averaged. One split-half mean was predicted from the other using a linear mixed-effects model with participant as a random intercept. The fixed-effect slope was interpreted as the within-person reliability estimate. A 1,000-iteration random split-half analysis was conducted as a sensitivity analysis.

17. Reference

Dudarev, V., Barral, O., Zhang, C., Davis, G., & Enns, J. T. (2023). On the reliability of wearable technology: A tutorial on measuring heart rate and heart rate variability in the wild. *Sensors*, 23, 5863.